

Group F

b) In this exercise we test the FD scheme. We run the programs using the data from Batches 1 to 4. Compare your answers with those from the previous exercises.

	T	K	sig	r	S	FDM C	FDM P	Given C	Given P	dC given	dP given
Batch 1	0.25	65	0.3	0.08	60	2.02564	5.738554	2.13337	5.84628	-0.10773	-0.10773
Batch 2	1	100	0.2	0	100	7.903546	7.903546	7.96557	7.96557	-0.06202	-0.06202
Batch 3	1	10	0.5	0.12	5	0.199154	4.068358	0.204058	4.07326	-0.0049	-0.0049
Batch 4	30	100	0.3	0.08	100	92.12137	1.193161	92.1757	1.2475	-0.05433	-0.05434

The FDM model does not give the exact answer like the BS formula because the Black–Scholes PDE is solved analytically, while FDM solves it numerically using approximations. In the code, the option price is computed on a finite difference grid defined by the spatial step h and the number of time steps N . Since derivatives are approximated with finite differences, discretization errors are introduced. Therefore, The accuracy depends on the grid parameters J (space steps) and N (time steps). Although `stableN()` ensures numerical stability, the solution remains an approximation, which is why small errors appear when comparing the FDM prices with the exact BS values (`givenCall`, `givenPut`). Increasing the number of time and space steps would reduce these errors.